[MP2] print_kmem_cache() 疑問

葉瞻語 (CHAN-YU YEH)

All Sections

5 5

助教您好，

我目前已完成 print_kmem_cache() 以外的implementation （如果將print_kmem_cache() 內容註解掉 testcases 都pass）

但是我在實作 print_kmem_cache() 時，遇到了以下問題：

我在 testcase mp2-0 的 output 中，跟 spec 範例 output 一模一樣（有 in-cache fragmentation optimization）

```
*** Now run 'gdb' in another window.

xv6 kernel is booting

[MP2] Slab size: 4
[FILE] fileinit
[SLAB] New kmem_cache (name: file, object size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, support in cache obj: 7) is created
hart 1 starting
hart 2 starting
[FILE] filealloc
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f59090 in slab 0x0000000087f59000 (file) is allocated and initialized
init: starting sh
[FILE] filealloc
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f59288 in slab 0x0000000087f59000 (file) is allocated and initialized
[FILE] fileclose
[SLAB] Free 0x0000000087f59288 in slab 0x0000000087f59000 (file)
[SLAB] End of free
$ mp2
[SLAB] kmem_cache { name: file, obj_size: 504, at: 0x0000000087f59000, in_cache_obj: 7 }
[SLAB]    [ cache slabs ]
[SLAB]       [ slab 0x0000000087f59000 ] { freelist: 0x0000000087f59288, nxt: 0x0000000000000000 }
[SLAB]          [ idx 0 ] { addr: 0x0000000087f59090, as_ptr: 0x0000000900000003, as_obj: {tp: 3, ref: 9, readable: 1, writable: 1, pipe: 0x0000000000000000, ip: 0x0000
[SLAB]          [ idx 1 ] { addr: 0x0000000087f59288, as_ptr: 0x0000000087f59480, as_obj: {tp: -2013948800, ref: 0, readable: 1, writable: 1, pipe: 0x0000000000000000,
[SLAB]          [ idx 2 ] { addr: 0x0000000087f59480, as_ptr: 0x0000000087f59678, as_obj: {tp: -2013948296, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505,
[SLAB]          [ idx 3 ] { addr: 0x0000000087f59678, as_ptr: 0x0000000087f59870, as_obj: {tp: -2013947792, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505,
[SLAB]          [ idx 4 ] { addr: 0x0000000087f59870, as_ptr: 0x0000000087f59a68, as_obj: {tp: -2013947288, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505,
[SLAB]          [ idx 5 ] { addr: 0x0000000087f59a68, as_ptr: 0x0000000087f59c60, as_obj: {tp: -2013946784, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505,
[SLAB]          [ idx 6 ] { addr: 0x0000000087f59c60, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 5, writable: 5, pipe: 0x0505050505050505, ip: 0x0505
[SLAB] print_kmem_cache end
$ qemu-system-riscv64: terminating on signal 15 from pid 65 (make)
```

但是 output 卻是

```
public/mp2-0: FAIL (0/3) (2.3s)
    Invalid first print data: SlabPrintfSlabTypeData(slab_type='cache')
    QEMU output saved to out/public/mp2-0.out.case
Score: 0/3
```

檢查多次都不知道是遇到了什麼問題，因此想請問助教看看 ~

謝謝助教！

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Reply



楊子慶 (YANG, TZU-CHING) (https://cool.ntu.edu.tw/courses/46296/users/229158)

Mar 24, 2025

同學您好：

這是因為在舊的 Spec 中 object_size 在列印說明處被誤植為 obj_size，導致同學輸出 obj_size 被直譯器判定為有誤。範例輸出則為 object_size，故能通過測試。

已在 Spec 中將錯誤修正並提交於 https://github.com/Shiritai/xv6-ntu-mp2/commit/e2dba993538ac25c765a8b117d0c06ae3d07094e (https://github.com/Shiritai/xv6-ntu-mp2/commit/e2dba993538ac25c765a8b117d0c06ae3d07094e)

建議同學利用以下方式下載新的文件：

https://cool.ntu.edu.tw/courses/46296/discussion_topics/382637 (https://cool.ntu.edu.tw/courses/46296/discussion_topics/382637)

若有更多問題，請不吝發問，感謝。

子慶



葉瞻語 (CHAN-YU YEH) (https://cool.ntu.edu.tw/courses/46296/users/76986)

Mar 24, 2025

改完後就沒問題了 ~

十分感謝助教！

Reply



林宗聖 (LIN, TZUNG-SHENG) (https://cool.ntu.edu.tw/courses/46296/users/125994)

Yesterday

借串問問大神類似的問題：

mp2-sunchigg > out > public > mp2-0.out.failed

```
3   xv6 kernel is booting
4
5   [MP2] Slab size: 4
6   [FILE] fileinit
7   [SLAB] New kmem_cache (name: file, object size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, support in cache obj: 0) is cre
8   hart 1 starting
9   hart 2 starting
10  [FILE] filealloc
11  [SLAB] Alloc request on cache file
12  [SLAB] Object 0x0000000087f54000 in slab 0x0000000087f4e000 (file) is allocated and initialized
13  (debug only) SLAB_BASE(obj): 0x0000000087f54000
14  init: starting sh
15  [FILE] filealloc
16  [SLAB] Alloc request on cache file
17  [SLAB] Object 0x0000000087f541f8 in slab 0x0000000087f4e000 (file) is allocated and initialized
18  (debug only) SLAB_BASE(obj): 0x0000000087f54000
19  [FILE] fileclose
20  [SLAB] Free 0x0000000087f541f8 in slab 0x0000000087f4e000 (file)
21  [SLAB] End of free
22  $ mp2
23  [SLAB] kmem_cache { name: file, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }
24  [SLAB]    [ partial slabs ]
25  [SLAB]        [ slab 0x0000000087f4e000 ] { freelist: 0x0000000087f541f8, next: 0x0000000000000000 }
26  [SLAB]           [ idx 0 ] { addr: 0x0000000087f541f8, as_ptr: 0x0000000087f543f0, as_obj: {tp: -2013969424, ref: 0, readable: 1, writable:
27  [SLAB]           [ idx 1 ] { addr: 0x0000000087f543f0, as_ptr: 0x0000000087f545e8, as_obj: {tp: -2013968920, ref: 0, readable: 5, writable:
28  [SLAB]           [ idx 2 ] { addr: 0x0000000087f545e8, as_ptr: 0x0000000087f547e0, as_obj: {tp: -2013968416, ref: 0, readable: 5, writable:
29  [SLAB]           [ idx 3 ] { addr: 0x0000000087f547e0, as_ptr: 0x0000000087f549d8, as_obj: {tp: -2013967912, ref: 0, readable: 5, writable:
30  [SLAB]           [ idx 4 ] { addr: 0x0000000087f549d8, as_ptr: 0x0000000087f54bd0, as_obj: {tp: -2013967408, ref: 0, readable: 5, writable:
31  [SLAB]           [ idx 5 ] { addr: 0x0000000087f54bd0, as_ptr: 0x0000000087f54dc8, as_obj: {tp: -2013966904, ref: 0, readable: 5, writable:
32  [SLAB]           [ idx 6 ] { addr: 0x0000000087f54dc8, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 5, writable: 5, pipe:
33  [SLAB] print_kmem_cache end
```

我的錯誤研究了好陣子還是找不到怎麼解決，
感覺從 Alloc之後就開始錯了，似乎應該要是下面這樣
[SLAB] Object 0x0000000087f54000 in slab **0x0000000087f59000** (file) is allocated and initialized
但我目前是[SLAB] Object 0x0000000087f54000 in slab 0x0000000087f4e000 (file) is allocated and initialized，就有下面了錯誤了
求問大神們

```
== Test public/mp2-0 (3%) ==
public/mp2-0: FAIL (0/3) (1.7s)
    Invalid SlabAllocMsg msg: Invalid obj_addr 0x87f54000 for slab 0x87f4e000, data: (SlabAllocRequestData(origin='Alloc request on
cache file', name='file'), SlabAllocObjData(addr=2280996864, origin='Object 0x0000000087f54000 in slab 0x0000000087f4e000 (file) is
allocated and initialized', name='file', slab_addr=2280972288))
    QEMU output saved to out/public/mp2-0.out.failed
Score: 0/3
```



彭詳睿 (PENG,XIANG-RUI) (https://cool.ntu.edu.tw/courses/46296/users/182168)

12:00am

有一個問題是，print_kmem_cache應該印出partial slab中全部的8個object，但同學只印出7個。注意到第一個被allocate的object是在0x0000000087f54000，但這個位置並沒有在print_kmem_cache時出現。

另外同學輸出[SLAB] Object 0x0000000087f54000 in slab **0x0000000087f4e000** (file) is allocated and initialized應該是正確的，因為你的object看起來是存在一個partial slab裡，所以<slab_addr>是0x0000000087f4e000沒錯~
如果是像葉同學有實作in cache memory，<slab_addr>才會是填0x0000000087f59000 (cache的記憶體位置)。



林宗聖 (LIN, TZUNG-SHENG) (https://cool.ntu.edu.tw/courses/46296/users/125994)

11:33am

我確實還沒實作in cache memory，感謝

謝謝彭同學點出我的問題：
1. print_kmem_cache應該印出partial slab中全部的8個object
2. 第一個被allocate的object是在0x0000000087f54000，但這個位置並沒有在print_kmem_cache時出現

```
[MP2] Slab size: 4
[FILE] fileinit
[SLAB] New kmem_cache (name: file, object size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, support in cache obj: 0) is cre
hart 2 starting
hart 1 starting
[FILE] filealloc
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f54000 in slab 0x0000000087f4e000 (file) is allocated and initialized
(debug only) SLAB_BASE(obj): 0x0000000087f54000
init: starting sh
[FILE] filealloc
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f541f8 in slab 0x0000000087f4e000 (file) is allocated and initialized
(debug only) SLAB_BASE(obj): 0x0000000087f54000
[FILE] fileclose
[SLAB] Free 0x0000000087f541f8 in slab 0x0000000087f4e000 (file)
[SLAB] End of free
$ mp2
[SLAB] kmem_cache { name: file, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }
[SLAB]    [ partial slabs ]
[SLAB]        [ slab 0x0000000087f4e000 ] { freelist: 0x0000000087f541f8, next: 0x0000000000000000 }
[SLAB]            [ idx 0 ] { addr: 0x0000000087f54000, as_ptr: 0x0000000900000003, as_obj: {tp: 3, ref: 9, readable: 1, writable: 1, pipe:
[SLAB]            [ idx 1 ] { addr: 0x0000000087f541f8, as_ptr: 0x0000000087f543f0, as_obj: {tp: -2013969424, ref: 0, readable: 1, writable:
[SLAB]            [ idx 2 ] { addr: 0x0000000087f543f0, as_ptr: 0x0000000087f545e8, as_obj: {tp: -2013968920, ref: 0, readable: 5, writable:
[SLAB]            [ idx 3 ] { addr: 0x0000000087f545e8, as_ptr: 0x0000000087f547e0, as_obj: {tp: -2013968416, ref: 0, readable: 5, writable:
[SLAB]            [ idx 4 ] { addr: 0x0000000087f547e0, as_ptr: 0x0000000087f549d8, as_obj: {tp: -2013967912, ref: 0, readable: 5, writable:
[SLAB]            [ idx 5 ] { addr: 0x0000000087f549d8, as_ptr: 0x0000000087f54bd0, as_obj: {tp: -2013967408, ref: 0, readable: 5, writable:
[SLAB]            [ idx 6 ] { addr: 0x0000000087f54bd0, as_ptr: 0x0000000087f54dc8, as_obj: {tp: -2013966904, ref: 0, readable: 5, writable:
[SLAB]            [ idx 7 ] { addr: 0x0000000087f54dc8, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 5, writable: 5, pipe:
[SLAB] print_kmem_cache end
```

我感覺應該都好了，但還是沒通過test，也發現其他test cases也都是一樣的錯誤ＱＱＱ

```
== Test public/mp2-0 (3%) ==
public/mp2-0: FAIL (0/3) (2.3s)
    Invalid SlabAllocMsg msg: Invalid obj_addr 0x87f54000 for slab 0x87f4e000, data: (SlabAllocRequestData(origin='Alloc request on
cache file', name='file'), SlabAllocObjData(addr=2280996864, origin='Object 0x0000000087f54000 in slab 0x0000000087f4e000 (file) is
allocated and initialized', name='file', slab_addr=2280972288))
    QEMU output saved to out/public/mp2-0.out.failed
Score: 0/3
```

↖ **Reply**